

Leaf Convergence Games: Partizan Games on Terminating Rooted Graphs

IE619 — Combinatorial Game Theory (2025)

Your Name / Presenter

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A new class of partizan games with guaranteed termination

Abstract

We introduce **Leaf Convergence Games (LCG)**, a partizan combinatorial game played on finite rooted graphs whose edges are labelled **L** (blue) or **R** (red), with the key structural constraint that **all maximal paths terminate at designated terminal nodes**.

A token starts at the root. **Left** moves along **L**-edges and **Right** along **R**-edges. The player unable to move loses (normal play).

Key properties:

- All games terminate in finite time (no cycles).
- Leaf structure completely determines endgame strategy.
- Rich game values: integers, switches, infinitesimals, dyadic rationals.

Introduction

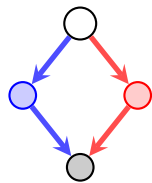
Leaf Convergence Games (LCG) are partizan games on finite rooted DAGs with the convergence property:

- Every maximal path from the root ends at a terminal node $\tau \in \mathcal{T}$.
- Edges labelled **L** (blue) or **R** (red).
- Token starts at root; Left moves on blue, Right on red.

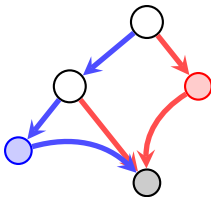
Why this class is interesting:

- **Termination guarantee** — always finite play.
- **Topology-agnostic** — linear, branched, multi-sink, arbitrary DAGs.
- **Strategic depth** — rich CGT values.

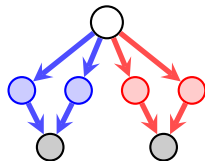
Motivating Structures



Linear convergence



Mixed depth



Multi-sink

Formal Framework

LCG Position:

- Finite DAG with root r and terminals $\mathcal{T}$.
- Edges labelled **L** or **R**.
- Every maximal path ends at a terminal.

Value:

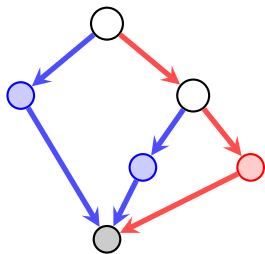
$$\mathcal{G}(v) = \{\mathcal{G}(c) : c \in \mathcal{L}(v) \mid \mathcal{G}(c) : c \in \mathcal{R}(v)\}$$

with $\mathcal{G}(\tau) = 0$.

Depth-1 Values

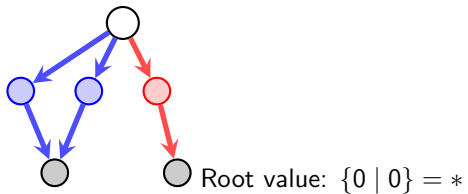
Left	Right	CGSuite	Value	Temp
1	0	$\{0\text{---}\}$	1	$-\infty$
2	0	$\{\{0\text{---}\}\text{---}\}$	2	$-\infty$
0	1	$\{\text{---}0\}$	-1	$-\infty$
1	1	$\{0\text{---}0\}$	*	0
1	2	$\{1\text{---}1\}$	± 1	1

Example 1: Asymmetric Depth Tree



Root value: $\{0 \mid *\} = \uparrow$

Example 2: Multi-Sink DAG



Computational Analysis

Python evaluator:

```
def evaluate_lcg(nodes, edges, root, terminals):
    memo = {}
    def eval_node(v):
        if v in terminals: return ([], [])
        left_opts = [eval_node(c) for s,c,col in edges if s==v and col
                        == 'L']
        right_opts = [eval_node(c) for s,c,col in edges if s==v and
                        col=='R']
        return (left_opts, right_opts)
    return eval_node(root)
```

Key Takeaways

- ① Termination is guaranteed by the convergence property.
- ② Topology (linear / branched / multi-sink) shapes the value.
- ③ Depth-2 already gives infinitesimals and switches.
- ④ Symmetry always gives value $*$.

Open Questions

- ① Closed-form characterisation for arbitrary DAGs.
- ② Complete enumeration of depth-3 values.
- ③ Efficient strategy computation for large positions.

References

 Berlekamp, Conway, Guy, *Winning Ways*.

 Conway, *On Numbers and Games*.

 Siegel, CGSuite.